



Impact of booster vaccination interval on SARS-CoV-2 infection, hospitalization, and death

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ABSTRACT

Objectives: We investigated how booster interval affects the risks of SARS-CoV-2 infection and Covid-19-related hospitalization and death in different age groups.

Methods: We collected data on booster receipts and Covid-19 outcomes between September 22, 2021 and February 9, 2023 for 5,769,205 North Carolina residents ≥ 12 years of age who had completed their primary vaccination series. We related Covid-19 outcomes to baseline characteristics and booster doses through Cox regression models.

Results: For adults ≥ 65 years of age, boosting every 9 months was associated with proportionate reductions (compared with no boosting) of 18.9% (95% CI, 18.5–19.4) in the cumulative frequency of infection, 37.8% (95% CI, 35.3–40.3) in the cumulative risk of hospitalization or death, and 40.9% (95% CI, 37.2–44.7) in the cumulative risk of death at 2 years after completion of primary vaccination. The reductions were lower by boosting every 12 months and higher by boosting every 6 months. The reductions were smaller for individuals 12–64 years of age.

Conclusion: Boosting at a shorter interval was associated with a greater reduction in Covid-19 outcomes, especially hospitalization and death. Frequent boosting conferred greater benefits for individuals aged ≥ 65 than for individuals aged 12–64.

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Introduction

Vaccination plays a very important role in preventing SARS-CoV-2 infection and severe Covid-19 disease. Primary vaccination was highly effective against infection, hospitalization, and death, but its effectiveness waned gradually over time and was further reduced by the emergence of new variants [1–4]. Booster vaccination restored some immunity, but the added protection declined rapidly over time, especially against newer variants [5–8]. Bivalent boosters were more effective than monovalent boosters, but their effectiveness also waned over time [9,10]. A question naturally arises: how often should booster doses be administered?

To answer this question, we examined North Carolina's statewide surveillance data on Covid-19 vaccination and disease incidence from the beginning of the booster campaign (September 22, 2021) to February 9, 2023 for 5,769,205 participants 12 years of age or older who had completed a primary vaccine series. We evaluated how booster interval affected the risks of SARS-CoV-2 infection, hospitalization, and death in two separate groups: individuals 65 years of age or older and individuals 12–64 years of age.

Methods

Background

On September 22, 2021, the US Food and Drug Administration (FDA) authorized the BNT162b2 (Pfizer-BioNTech) vaccine for emergency use as a single booster dose, to be given at least 6 months after completion of the primary series in individuals 65 years of age or older and in individuals 18–64 years of age at high risk of severe Covid-19 [11]. The authorization was extended to

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the mRNA-1273 (Moderna) and Ad26.COV2.S (Johnson & Johnson–Janssen) vaccines on October 20, 2021, to all individuals 18 years of age or older on November 19, 2021, and to children as young as 12 years old on February 9, 2022 [12–14].

Data sources

The North Carolina Covid-19 Surveillance System (NC COVID) is a central repository of person-level investigational data on Covid-19 cases and clinical outcomes. Clinical laboratories are mandated to report positive SARS-CoV-2 test results. The database contains all positive SARS-CoV-2 test results for each person using a unique identifier. Covid-19 related hospitalizations and deaths are determined by local health departments through follow-up investigations of Covid-19 cases [3,7,9,10]. Starting on January 1 of 2022, vital records have been used to improve the coverage and accuracy of Covid-19 death reporting.

The cloud-based Covid-19 Vaccine Management System (CVMS) tracked provider enrollment and vaccine product administration and allowed the state to manage vaccine supply. CVMS records were transferred daily to the North Carolina Department of Health and Human Services Business Intelligence Data Platform, where the data were processed to generate a recipient-based view of Covid-19 vaccination history.

Study design

We linked the NC COVID and CVMS databases through a Master Patient Index, and we extracted individual-level data on vaccination histories, as well as on SARS-CoV-2 infection, hospitalization, and death, between the booster authorization date of September 22, 2021 and the study end date of February 9, 2023 for all participants who had completed a primary series. We limited the study to mRNA boosters by excluding participants who received non-mRNA boosters. We then evaluated the impact of booster interval on the risks of SARS-CoV-2 infection, hospitalization, and death for two major age groups: 12–64 years old versus ≥ 65 years old.

Statistical analysis

We considered three types of outcomes: (i) SARS-CoV-2 infection, (ii) severe SARS-CoV-2 infection (i.e., SARS-CoV-2 infection that results in hospitalization or death), and (iii) death. The second outcome is a composite endpoint of hospitalization and death, which is dominated by hospitalization because death typically occurs after hospitalization. (It is difficult to consider hospitalization by itself because death is a competing risk for hospitalization.) We treated SARS-CoV-2 infection as a recurrent event and treated severe SARS-CoV-2 infection and death as single events.

For each of the two age groups, we fit a proportional rates model [15] to data on times to recurrent SARS-CoV-2 infections, a proportional hazards model [16] to data on time to severe infection, and a proportional hazards model to data on time to death. (Positive tests at least 90 days apart were considered recurrent infections.) Each model included the baseline characteristics listed in Table 1 (age, sex, race/ethnicity, socioeconomic status, geographic region, county-level vaccination rate, manufacturer of primary series, time period of the completion of primary series) as time-independent covariates. Booster doses were included as time-dependent covariates; the effect of a booster dose was allowed to depend nonparametrically on the type of vaccine (monovalent versus bivalent), the booster number (first, second, or third), and on the date of receipt of the booster dose, and the log rate ratio or hazard ratio for the booster effect was represented by a piecewise linear function of time since receipt of the booster dose, such that there was a separate function for each combination of the vaccine

type, the booster number, and the time period of booster receipt. The model also included infection history as a time-dependent covariate, whose effect was allowed to depend on the variant (pre-delta, delta, or omicron), and the log rate ratio or hazard ratio was represented by a linear function of time 2 weeks after infection. The effects of booster doses and previous infections were additive on the log scale. We let certain parameters be the same to ensure stable estimates. We estimated the parameters in each model by maximizing the partial likelihood with potentially censored observations [15,16].

We predicted the occurrences of infection, severe infection, and death under various booster intervals using Monte Carlo simulation. Specifically, we simulated times to recurrent infections, time to severe infection, and time to death from the fitted models by assigning each person to the same booster interval (e.g., one dose every 6 months) while fixing their baseline covariates at the observed values. We then estimated the cumulative frequency of infection (mean number of infections per person) by the Nelson-Aalen estimator [17] and the cumulative risks of severe infection and death by the Kaplan-Meier estimator [18]. We considered 4 booster intervals: no booster, one dose every 6 months, one dose every 9 months, and one dose every 12 months. For each booster interval, we repeated the aforementioned simulation procedure 100 times and averaged the resulting 100 sets of estimates. In addition, we estimated the risk difference by subtracting the cumulative frequency or risk for each booster interval from that of no booster. Of note, the inverse of the risk difference for a booster interval is the number of individuals needed to be boosted at that interval in order to prevent one adverse event. Finally, we estimated the vaccine effectiveness over time by one minus the ratio of the cumulative frequency or risk for each booster interval to that of no booster. We constructed the confidence intervals for each function by estimating the standard error through nonparametric bootstrap with 50 bootstrap samples.

A more detailed description of the statistical analysis is given in Supplemental Methods. The computer code is available at <https://dlin.web.unc.edu/software/booster-interval/>.

Results

Study population

Table 1 summarizes the demographic characteristics, booster doses administered, and clinical outcomes of the study participants. Among the 4,231,784 participants aged 12–64 who had completed a primary series, 2,140,349 received at least one booster dose; among the 1,537,421 participants aged 65 or older who had completed a primary series, 1,151,153 received at least one booster dose. The first booster doses were mostly monovalent vaccines, whereas the second and third booster doses were mainly bivalent vaccines. Among participants 12–64 years of age, there were 785,188 infections, 2817 severe infections, and 724 deaths; 401,826 (51.2%) of the infections, 1552 (55.1%) of the severe infections, and 529 (73.1%) of the deaths occurred in the 2,091,435 participants who had never been boosted. Among participants 65 years of age or older, there were 197,430 infections, 7302 severe infections, and 3630 deaths; 55,449 (28.1%) of the infections, 3394 (46.5%) of the severe infections, and 2272 (62.6%) of the deaths occurred in the 386,268 participants who had never been boosted.

Effects of booster interval on COVID-19 outcomes

Tables S1–S6 and Figures S1–S6 show the estimation results for the proportional rates models with recurrent SARS-CoV-2 infections and the proportional hazards models with severe SARS-CoV-2 infection and death in the two age groups. The fitted models

Table 1
Demographic and Clinical Characteristics of Study Participants.

Characteristic	12-64 Years of Age				65 Years of Age or Older			
	No. of Persons	Infection No. of Events	Hospitalization	Death	No. of Persons	Infection No. of Events	Hospitalization	Death
Total	4,231,784	785,188	2817	724	1,537,421	197,430	7302	3630
Booster doses								
No booster	2,091,435	401,826	1552	529	386,268	55,449	3394	2272
Monovalent vaccine								
First booster	1,983,459	360,084	1163	194	1,092,706	136,067	3743	1350
Second booster	131,015	19,451	72	8	243,245	29,527	636	165
Third booster	122	17	0	0	25	3	0	0
Bivalent vaccine								
First booster	156,890	23,278	102	1	58,447	5914	165	8
Second booster	454,360	81,467	220	4	313,615	38,899	653	36
Third booster	70,024	9746	33	1	155,995	17,757	293	27
Age group								
12-17	287,749	40,487	53	1				
18-29	813,870	160,063	146	6				
30-39	779,459	157,860	236	20				
40-49	816,652	164,349	388	92				
50-64	1,534,054	262,429	1994	605				
65-74					784,373	99,891	2125	804
≥75					753,048	97,539	5177	2826
Socioeconomic status								
Low	1,879,391	357,568	1772	429	848,255	114,847	4562	2203
High	2,352,393	427,620	1045	295	689,166	82,583	2740	1427
Sex								
Female	2,272,630	484,160	1582	373	873,630	111,926	3526	1639
Male	1,959,154	301,028	1235	351	663,791	85,504	3776	1991
Race/ethnicity								
Black	871,818	177,619	912	225	258,003	34,576	1302	610
Non-Black	3,359,966	607,569	1905	499	1,279,418	162,854	6000	3020
Geographic region								
Coastal Plain	961,136	181,281	835	221	434,094	59,299	2010	1048
Piedmont	2,864,223	539,105	1753	424	881,766	114,186	4665	2081
Mountain	406,425	64,802	229	79	221,561	23,945	627	501
County vaccination rate								
<62%	1,036,648	199,969	1201	298	474,307	66,811	2953	1391
62-75%	1,292,264	232,461	1004	245	571,005	72,124	2785	1377
>75%	1,902,872	352,758	612	181	492,109	58,495	1564	862
Primary series date								
12/10/2020-03/10/2021	403,816	103,879	274	64	812,865	111,842	3748	1887
03/11/2021-06/08/2021	2,439,303	461,617	1698	474	577,515	70,359	2878	1381
06/09/2021-09/06/2021	682,467	119,955	453	98	64,351	7986	394	205
09/07/2021-12/05/2021	461,981	79,162	297	77	43,138	5095	224	133
12/06/2021-03/05/2022	164,689	16,390	77	10	25,017	1509	39	20
03/06/2022-06/03/2022	60,465	3423	17	1	13,004	557	17	3
06/04/2022-02/09/2023	19,063	762	1	0	1531	82	2	1
Primary series manufacturer								
mRNA-1273	1,396,761	246,113	1109	296	744,318	91,968	3292	1735
BNT162b2	2,496,037	474,579	1348	338	749,420	99,088	3641	1686
Ad26.COV2.S	338,986	64,496	360	90	43,683	6374	369	209

were used to generate the event times under each booster interval. Figure 1 shows the cumulative frequency of infection and the cumulative risks of severe infection and death under no boosting or each of the 3 booster intervals, separately for the two age groups. Figures 2 and 3 show the corresponding risk differences and vaccine effectiveness. Table 2 shows the number of individuals needed to be boosted at a certain interval in order to prevent one adverse event within 2 years.

SARS-CoV-2 infection in participants 12-64 years of age

The cumulative frequencies of infection at 2 years after completion of a primary series were 0.201 (95% CI, 0.200-0.201), 0.190 (95% CI, 0.189-0.191), 0.172 (95% CI, 0.172-0.173), and 0.161 (95% CI, 0.159-0.163) under no boosting and boosting every 12 months, every 9 months, and every 6 months, respectively (Figure 1A, Table S7). The risk differences were 0.011 (95% CI, 0.010-0.011), 0.028 (95% CI, 0.028-0.029), and 0.040 (95% CI, 0.038-0.042) under boosting every 12 months, every 9 months, and every 6 months,

respectively (Figure 2A, Table S8), and the corresponding vaccine effectiveness was 5.4% (95% CI, 5.1-5.6), 14.1% (95% CI, 13.9-14.3), and 19.8% (95% CI, 18.8-20.9), respectively (Figure 3A, Table S9).

Severe SARS-CoV-2 infection in participants 12-64 years of age

The cumulative risk of severe infection at 2 years after completion of a primary series was 9.2×10^{-4} (95% CI, 8.7-9.7), 8.5×10^{-4} (95% CI, 8.0-9.0), 6.9×10^{-4} (95% CI, 6.6-7.2), or 6.2×10^{-4} (95% CI, 5.6-6.8) under no boosting, boosting every 12 months, boosting every 9 months, or boosting every 6 months, respectively (Figure 1C, Table S7). The risk difference was 0.74×10^{-4} (95% CI, 0.36-1.11), 2.34×10^{-4} (95% CI, 1.91-2.78), or 3.07×10^{-4} (95% CI, 2.22-3.91) under boosting every 12 months, every 9 months, or every 6 months, respectively (Figure 2C, Table S8), and the corresponding vaccine effectiveness was 8.0% (95% CI, 4.0-11.9), 25.4% (95% CI, 21.7-29.0), or 33.2% (95% CI, 25.0-41.4), respectively (Figure 3C, Table S9).

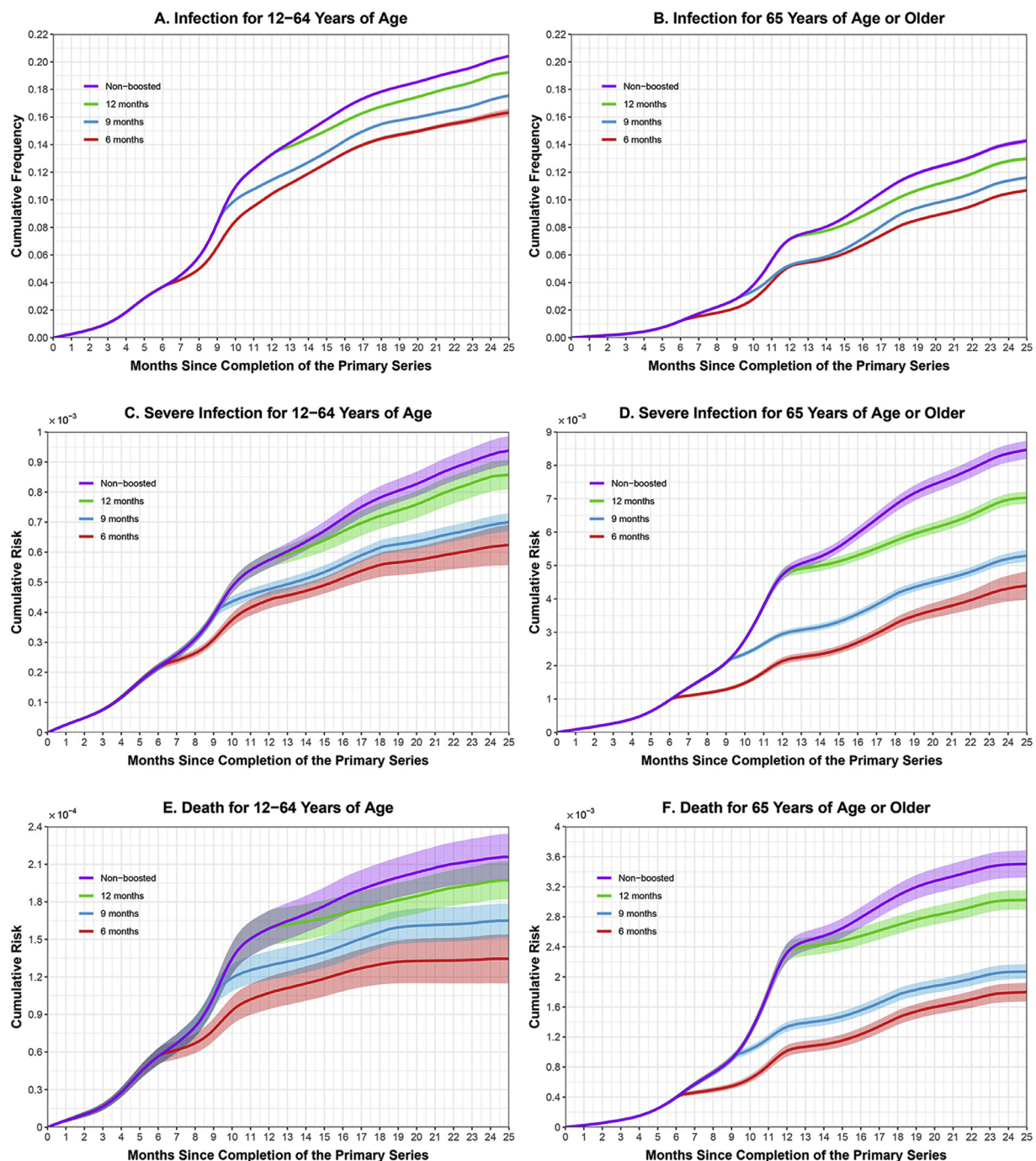


Figure 1. Cumulative frequencies of SARS-CoV-2 infection, cumulative risks of severe SARS-CoV-2 infection resulting in hospitalization or death, and cumulative risks of Covid-19-related death under no booster and three booster intervals, as a function of time since completion of a primary series, by age group. Estimates are shown by solid curves, and 95% confidence intervals based on bootstrap standard error estimates are shown by shaded bands.

Death in participants 12–64 years of age

The cumulative risk of death at 2 years after completion of a primary series was 2.15×10^{-4} (95% CI, 1.97–2.32), 1.96×10^{-4} (95% CI, 1.82–2.11), 1.64×10^{-4} (95% CI, 1.51–1.78), or 1.34×10^{-4} (95% CI, 1.13–1.56) under no boosting, boosting every 12 months, boosting every 9 months, or boosting every 6 months, respectively (Figure 1E, Table S7). The risk difference was 1.86×10^{-5} (95% CI, 1.12–2.61), 5.04×10^{-5} (95% CI, 3.65–6.43), or 8.05×10^{-5} (95% CI, 5.75–10.3) under boosting every 12 months, every 9 months, or every 6 months, respectively (Figure 2E, Table S8), and the corresponding vaccine effectiveness was 8.7% (95% CI, 5.5–11.8), 23.5% (95% CI, 17.9–29.0) or 37.5% (95% CI, 27.5–47.4), respectively

(Figure 3E, Table S9). The number of Individuals needed to be boosted every 12 months, every 9 months, or every 6 months in order to prevent one death within 2 years was 53,671 (95% CI, 38,342–89,421), 19,842 (95% CI, 15,561–27,374), or 12,426 (95% CI, 9668–17,384), respectively (Table 2).

SARS-CoV-2 infection in participants 65 years of age or older

The cumulative frequencies of infection at 2 years after completion of a primary series were 0.141 (95% CI, 0.140–0.141), 0.128 (95% CI, 0.127–0.129), 0.114 (95% CI, 0.113–0.115), and 0.105 (95% CI, 0.104–0.105) under no boosting and boosting every 12 months, every 9 months, and every 6 months, respectively (Figure 1B, Table

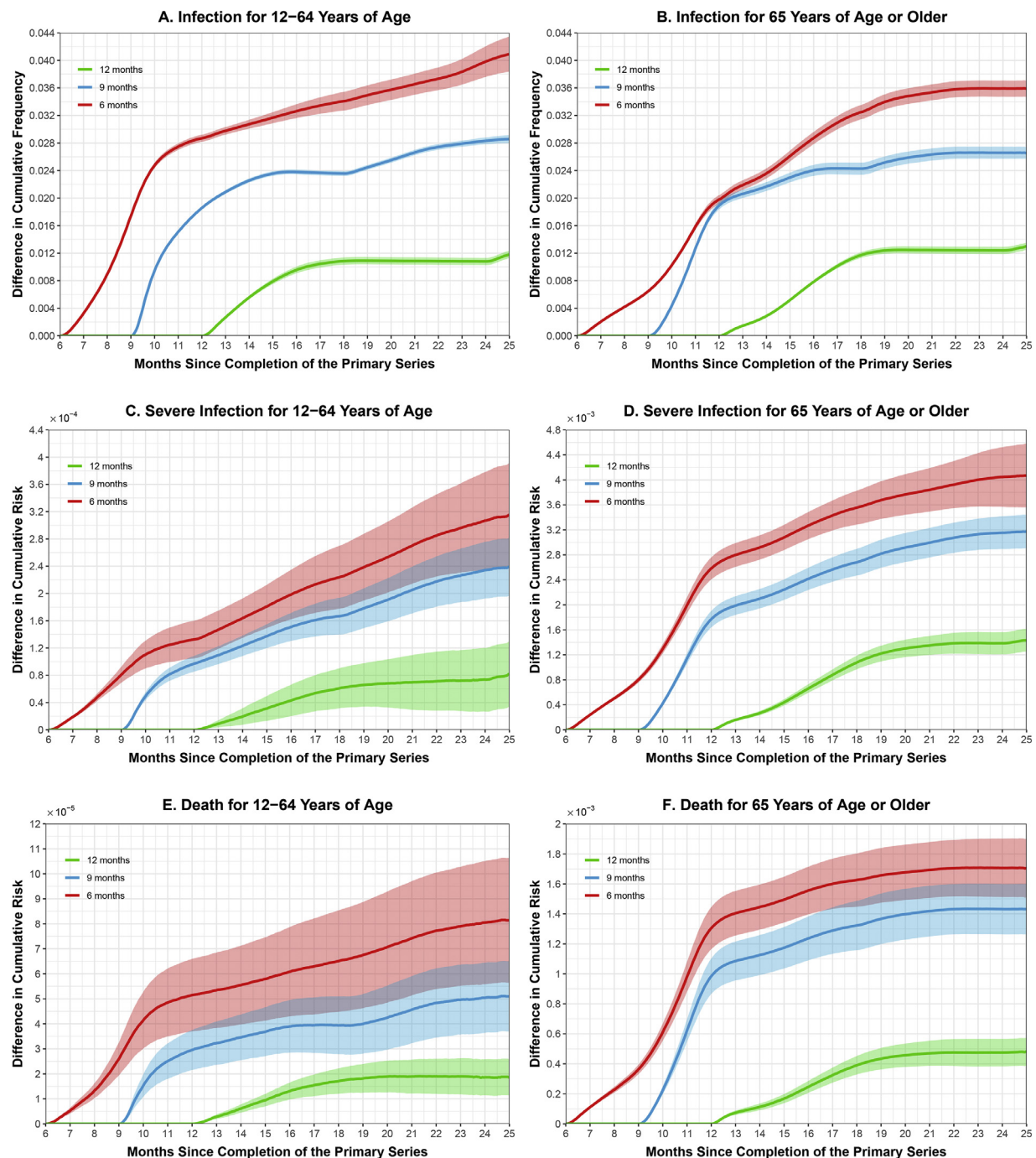


Figure 2. Risk differences in SARS-CoV-2 infection, severe SARS-CoV-2 infection, and Covid-19-related death for three booster intervals (compared to no booster), as a function of time since completion of a primary series, by age group. Estimates are shown by solid curves, and 95% confidence intervals based on bootstrap standard error estimates are shown by shaded bands.

S7). The risk differences were 0.012 (95% CI, 0.012–0.013), 0.027 (95% CI, 0.026–0.027), and 0.036 (95% CI, 0.035–0.037) under boosting every 12 months, every 9 months, and every 6 months, respectively (Figure 2B, Table S8), and the corresponding vaccine effectiveness was 8.8% (95% CI, 8.5–9.1), 18.9% (95% CI, 18.5–19.4), and 25.6% (95% CI, 24.9–26.2), respectively (Figure 3B, Table S9).

Severe SARS-CoV-2 infection in participants 65 years of age or older

The cumulative risk of severe infection at 2 years after completion of a primary series was 8.4×10^{-3} (95% CI, 8.1–8.6), 7.0×10^{-3} (95% CI 6.8–7.2), 5.2×10^{-3} (95% CI, 5.0–5.4), or 4.3×10^{-3} (95% CI, 3.9–4.8) under no boosting, boosting every 12 months, boosting every

every 9 months, or boosting every 6 months, respectively (Figure 1D, Table S7). The risk difference was 1.39×10^{-3} (95% CI, 1.21–1.56), 3.16×10^{-3} (95% CI, 2.87–3.44), or 4.05×10^{-3} (95% CI, 3.48–4.61) under boosting every 12 months, every 9 months, or every 6 months, respectively (Figure 2D, Table S8), and the corresponding vaccine effectiveness was 16.6% (95% CI, 14.9–18.3), 37.8% (95% CI, 35.3–40.3), or 48.4% (95% CI, 42.6–54.2), respectively (Figure 3D, Table S9).

Death in participants 65 years of age or older

The cumulative risk of death at 2 years after completion of a primary series was 3.50×10^{-3} (95% CI, 3.29–3.70), 3.02×10^{-3}

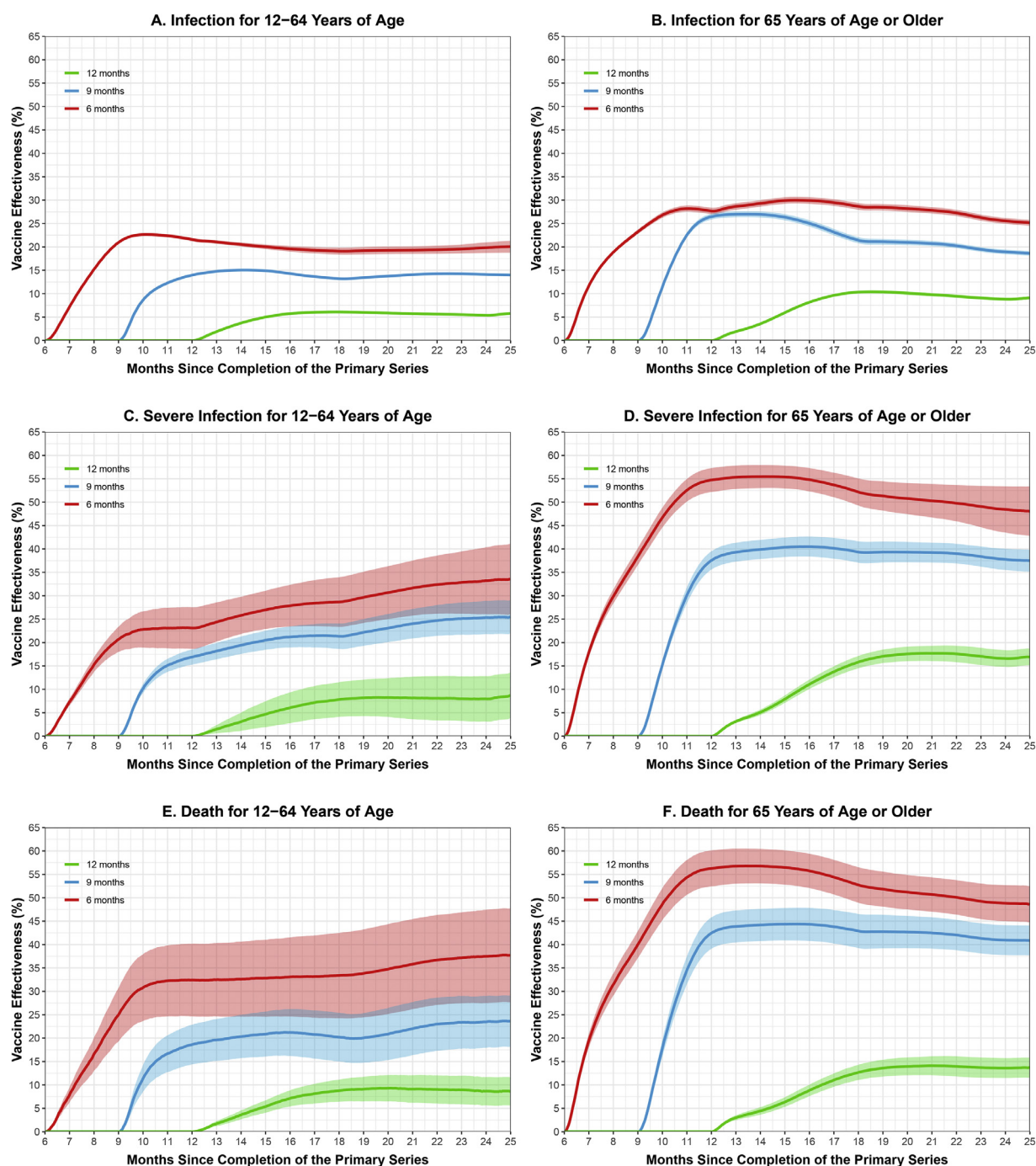


Figure 3. Vaccine effectiveness (%) against SARS-CoV-2 infection, Severe SARS-CoV-2 infection, and Covid-19-related death for three booster intervals (compared to no booster), as a function of time since completion of a primary series, by age group. Vaccine effectiveness is defined as one minus the ratio of the cumulative frequency or risk for a specific booster interval to that of no booster. Estimates are shown by solid curves, and 95% confidence intervals based on bootstrap standard error estimates are shown by shaded bands.

(95% CI, 2.87–3.18), 2.07×10^{-3} (95% CI, 1.96–2.17), or 1.79×10^{-3} (95% CI, 1.66–1.92) under no boosting, boosting every 12 months, boosting every 9 months, or boosting every 6 months, respectively (Figure 1F, Table S7). The risk difference was 0.48×10^{-3} (95% CI, 0.39–0.56), 1.43×10^{-3} (95% CI, 1.24–1.63), 1.71×10^{-3} (95% CI, 1.48–1.93) under boosting every 12 months, every 9 months, or every 6 months, respectively (Figure 2F, Table S8), and the corresponding vaccine effectiveness was 13.6% (95% CI, 11.6–15.6), 40.9% (95% CI, 37.2–44.7), or 48.8% (95% CI, 44.4–53.2), respectively (Figure 3F, Table S9). The number of Individuals needed to be boosted every 12 months, every 9 months, or every 6 months

in order to prevent one death within 2 years would be 2105 (95% CI, 1780–2574), 699 (95% CI, 615–809), or 586 (95% CI, 517–676), respectively (Table 2).

Discussion

Boosting at shorter intervals was found to be associated with lower risks of SARS-CoV-2 infection, hospitalization, and death. The effectiveness of each booster interval was higher against hospitalization and death than against infection, reflecting the fact that

Table 2

Estimates and 95% Confidence Intervals for the Number of Individuals Needed to be Boosted at a Certain Interval in Order to Prevent One Adverse Event Within 2 Years.

12–64 Years of Age			
Booster Interval	Infection	Severe Infection	Death
Every 12 months	93 (89–97)	13,570 (8983–27,729)	53,671 (38,342–89,421)
Every 9 months	35 (35–36)	4266 (3599–5236)	19,842 (15,561–27,374)
Every 6 months	25 (24–27)	3259 (2555–4500)	12,426 (9668–17,384)
65 Years of Age or Older			
Booster Interval	Infection	Severe Infection	Death
Every 12 months	81 (78–84)	722 (640–827)	2105 (1780–2574)
Every 9 months	38 (37–39)	317 (290–349)	699 (615–809)
Every 6 months	28 (27–29)	247 (217–298)	586 (517–676)

boosters were more effective and more durable against hospitalization and death than against infection [7,10].

The effectiveness of each booster interval was greater for adults 65 years of age or older than individuals 12–64 years of age, especially against hospitalization and death. In addition, the risks of hospitalization and death were much higher in the older age group than in the younger age group. Thus, frequent boosting would confer greater benefits for the older age group than the younger age group.

Because the risk of death from Covid-19 was relatively low among individuals 12–64 years of age, a very large number of doses would be required under any booster interval in order to prevent one Covid-19 death in that age group. For adults 65 years of age or older, boosting every 9 months would require a smaller number of doses than boosting every 6 months or every 12 months in order to prevent one death.

On January 26, 2023, the FDA proposed an annual Covid-19 vaccination plan, in which a two-dose series will be given to certain young children and to older adults and persons with compromised immunity, and only one dose will be given to all other individuals every fall. On June 6, 2023, the European Centre for Disease Prevention and Control and the European Medicines Agency issued a joint statement that, for the upcoming fall vaccination campaigns, a priority should be targeting individuals that are at risk of progression to severe disease once infected because of risk factors, such as older adults, immunocompromised individuals, those with underlying medical conditions irrespective of age, and pregnant women. The findings from our study broadly support these plans.

The effects of a booster dose might depend on the type of vaccine, the booster number, the circulating variant, time since receipt of the booster dose, and infection history. Even with nearly 6 million participants, however, there were not enough data to reliably estimate a very large number of parameters. Thus, we did not distinguish between the Moderna and Pfizer-BioNTech boosters. In addition, we assumed no interactions between booster history and infection history. However, our previous work suggested that the assumptions we made were good first-order approximations [7,9,10].

The FDA expanded eligibility for the BNT162b2 vaccine booster dose to children 5–11 years of age on May 17, 2022 and authorized bivalent vaccines as booster doses for children as young as 6 months on December 8, 2022 [19,20]. Not enough time has passed to assess the effects of booster interval on clinical outcomes among children under the age of 12 years.

It could take years to amass sufficient clinical trial data to inform optimal booster intervals. Surveillance data provide a valuable resource to generate the needed evidence. Even contingent upon completion of primary vaccination, however, there is a possibility of selection bias regarding who does and does not receive

a booster. We adjusted for potential confounders, such as demographic variables, socioeconomic status, and geographical location, in our models. In addition, we avoided confounding due to time-varying factors (e.g., level of community transmission and prevalence of new variants) by measuring all event times from the start of the study and thus comparing boosted and non-boosted individuals on the same calendar date.

This study was limited by unmeasured confounding bias. In particular, we did not have information on underlying medical conditions (immunosuppression status, cardiovascular diseases, respiratory diseases) which might have increased vaccine uptakes and also increased the risks of infection, hospitalization, and death. Indeed, the third dose was part of the mRNA primary series for immunocompromised individuals rather than a booster dose. We were unable to separate additional doses for immunocompromised individuals from booster doses, such that our estimates of booster effectiveness might have been attenuated.

At-home SARS-CoV-2 tests became more common over the course of this study. Our database did not include home-based testing results, so cases of SARS-CoV-2 infection were underdocumented, especially during the latter part of the study. In addition, local health departments were unable to follow many Covid-19 cases to ascertain their hospitalization and survival status. Thus, our estimates of the cumulative frequency of infection and the cumulative risks of hospitalization and death were biased downward. However, the estimates of frequency ratios and risk ratios were less sensitive to underreporting, provided that the outcome ascertainment rate did not depend markedly on booster interval. Since January 1, 2022, the North Carolina Division of Public Health has used vital records as an additional means to identify Covid-19 deaths, such that it was unlikely for Covid-19 deaths to go unreported. Therefore, our conclusions about the effectiveness of different booster intervals should be valid, especially for the comparisons between the two age groups and for the endpoint of death.

Data sharing statement

Deidentified individual-level participant data and data dictionary will be available for sharing after approval of a proposal by the North Carolina Department of Public Health and following a signed data access agreement. Requests for data sharing can be directed to Dr. Sunny.

Declaration of competing interest

All authors declare no competing interests.

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Ethical approval

The study was approved by the institutional review board. The requirement for written informed consent was waived because the data were anonymous.

Author contribution

DYL designed the study, supervised data analysis, and prepared the manuscript. YX and YG performed data analysis and prepared figures and tables. DZ supervised data analysis. ZM and SKS collected data and contributed to data interpretation. SKS and YX directly accessed and verified the data. DYL decided to submit the manuscript for publication. All authors reviewed and approved the final text.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijid.2024.107084](https://doi.org/10.1016/j.ijid.2024.107084).

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